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COMPUTER

CHAPTER 15

Operating System

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COMPUTER · CHAPTER 15

Operating System

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IN THIS CHAPTER

- Computer System Components
- Components Of Operating System
- Functions Of Operating System
- Types Of Operating System
- System Call
- Most Important Facts For Exams

An **Operating System (OS)** is system software that acts as an intermediary between the user and the computer hardware.

It is a collection of system programs that controls and coordinates all operations of a computer system.

Operating System = Interface between User and Hardware

Example

When you open a browser, play music, print a document, or save a file, the Operating System manages all these activities.

EXAMPLES OF OPERATING SYSTEMS

- MS-DOS
- Microsoft Windows
- Windows NT
- UNIX
- Linux
- macOS
- OS/2
- VMS
- MVS
- VM

Modern Examples

- Windows 10
- Windows 11
- Ubuntu Linux
- Android
- iOS
- macOS

COMPUTER SYSTEM COMPONENTS

A computer system consists of four major components:

Computer System

│

├── Hardware

├── Operating System

├── Application Programs

└── Users

1. Hardware

Provides basic computing resources.

Examples

- CPU
- Memory
- Keyboard
- Mouse
- Monitor
- Hard Disk

2. Operating System

Controls and coordinates the use of hardware resources.

Functions

- Manages CPU
- Manages Memory
- Manages Files
- Manages Devices

3. Application Programs

Programs that help users perform specific tasks.

Examples

- MS Word
- Web Browser
- Video Games
- Database Software
- Compiler

4. Users

Entities that use the computer.

Examples

- Humans
- Machines
- Other Computers

COMPONENTS OF OPERATING SYSTEM

Operating System mainly consists of:

1. KERNEL

Kernel is the active core component of the Operating System.

It directly interacts with computer hardware.

Characteristics

- Runs continuously
- Manages hardware resources
- Controls CPU, memory, and devices

Examples

- Device Drivers
- System Files
- DLL Files

Functions

- Process Management
- Memory Management
- Device Management
- File Management

Exam Point

Kernel interacts directly with hardware.

2. SHELL

Shell is the command interpreter of the Operating System.

It acts as an interface between:

- User
- Operating System

Functions

- Accepts commands from users
- Interprets commands
- Passes commands to the Kernel

Examples

- Command Prompt (Windows)
- Bash Shell (Linux)

Exam Point

Shell = User Interface of OS

Feature	Kernel	Shell
Position	Core of OS	Outer Interface
Interaction	Hardware	User
Function	Resource Management	Command Interpretation
Runs Continuously	Yes	No

FUNCTIONS OF OPERATING SYSTEM

Functions of OS



│

├── Process Management

├── Memory Management

├── File Management

├── I/O Management

└── Secondary Storage Management

1. PROCESS MANAGEMENT

A Process is a program that is currently executing.

Resources Needed by a Process

- CPU Time
- Memory
- Files
- I/O Devices

2. MAIN MEMORY MANAGEMENT

Main Memory (RAM) stores data and instructions currently being used by the CPU.

3. FILE MANAGEMENT

A File is a collection of related information.

Examples

- Text Files
- Programs
- Images
- Databases

Functions of OS

- File Creation
- File Deletion
- Directory Creation
- Directory Deletion

- File Backup
- Storage Mapping

4. INPUT/OUTPUT (I/O) MANAGEMENT

I/O Management controls communication between computer and peripheral devices.

Examples

- Printer Driver
- Scanner Driver
- Sound Driver

Exam Point

Device Driver acts as translator between OS and hardware.

5. SECONDARY STORAGE MANAGEMENT

Secondary Storage stores data permanently.

Examples

- Hard Disk
- SSD
- CD/DVD
- Pen Drive

TYPES OF OPERATING SYSTEM

Operating Systems

│

├── Batch OS

├── Multiprogramming OS

├── Multiprocessing OS

├── Multitasking OS

├── Network OS

├── Real-Time OS

├── Time-Sharing OS

└── Embedded OS

1. BATCH OPERATING SYSTEM

Jobs of similar type are collected into batches and executed one by one.

Characteristics

- No direct interaction with user
- Uses FCFS (First Come First Serve) Scheduling

2. MULTIPROGRAMMING OPERATING SYSTEM

Multiple programs remain in memory simultaneously.

When one program performs I/O operations, CPU executes another program.

3. MULTIPROCESSING OPERATING SYSTEM

Uses two or more processors simultaneously.

Characteristics

- Parallel Processing
- Higher Throughput
- Faster Execution

Example

Supercomputers

4. MULTITASKING OPERATING SYSTEM

Allows a user to run multiple tasks simultaneously.

Example

Using:

- MS Word
- Browser
- Music Player at the same time.

5. NETWORK OPERATING SYSTEM (NOS)

An OS that supports communication between computers over a network.

Examples

- Windows Server
- UNIX
- Linux Server

6. REAL-TIME OPERATING SYSTEM (RTOS)

A Real-Time Operating System guarantees that tasks are completed within a specified deadline.

Applications

- Missile Systems
- Air Traffic Control
- Industrial Automation
- Medical Monitoring Systems
- Weather Monitoring Systems

7. TIME-SHARING OPERATING SYSTEM

CPU time is divided among multiple users and programs.

Each user gets a small time slice.

Characteristics

- Multi-user system
- Interactive processing

Exam Point

Time-Sharing = Many Users Share CPU Time

8. EMBEDDED OPERATING SYSTEM

An Embedded OS is designed for dedicated devices.

Examples

- Smart Watches
- Microwave Ovens
- Washing Machines
- Refrigerators
- Smart TVs

Exam Point

Embedded OS is used in Smart Devices.

SYSTEM CALL

A System Call is a request made by a program to the Operating System for a service.

System Call = Interface between Process and OS

TYPES OF SYSTEM CALLS

1. Process Control

Functions

- Create Process
- Terminate Process
- Load Process

2. Device Management

Functions

- Read Device
- Write Device
- Device Control

3. File Management

Functions

- Create File
- Delete File
- Open File
- Close File

MOST IMPORTANT FACTS FOR EXAMS

1. Operating System acts as an interface between user and hardware.

2. Kernel is the core of OS.

3. Shell is the command interpreter.

4. Process is a program in execution.

5. RAM is volatile memory.

6. Hard Disk is non-volatile memory.

7. Batch OS uses FCFS scheduling.

8. Multiprogramming improves CPU utilization.

9. Multiprocessing uses multiple processors.

10. Multitasking allows multiple tasks simultaneously.

11. RTOS works under strict deadlines.

12. Embedded OS is used in smart devices.

13. System Call is an interface between process and OS.

14. Device Drivers help OS communicate with hardware.

15. Time-Sharing OS supports multiple users.

PROCESS MANAGEMENT IN OPERATING SYSTEM

A **Program** is a set of instructions (commands written in code) stored in memory.

A program does nothing until its instructions are executed by the CPU.

A **Process** is a program that is currently being executed.

Process = Program in Execution

Examples

Program	Process
MS Word installed on disk	MS Word currently running
Chrome Browser	Chrome opened and executing
Media Player	Playing a video

WHY PROCESS MANAGEMENT IS NEEDED?

In a computer system:

- Multiple processes run simultaneously.
- Many processes may need the same resources.
- Resources are limited.

Resources Required by Processes

- CPU
- Memory (RAM)
- Files
- Printers
- Input/Output Devices

The Operating System manages these resources efficiently to avoid conflicts.

Important Fact

If multiple processes try to use the same resource simultaneously without proper management, the system may enter a **Deadlock State**.

Exam Point

Process Management prevents resource conflicts and improves CPU utilization.

PROCESS LIFE CYCLE

A process passes through several states from creation to termination.

New

|

▼

Ready

/ \

▼ ▼

Running Suspend Ready

/ \

▼ ▼

Waiting Terminated

|

▼

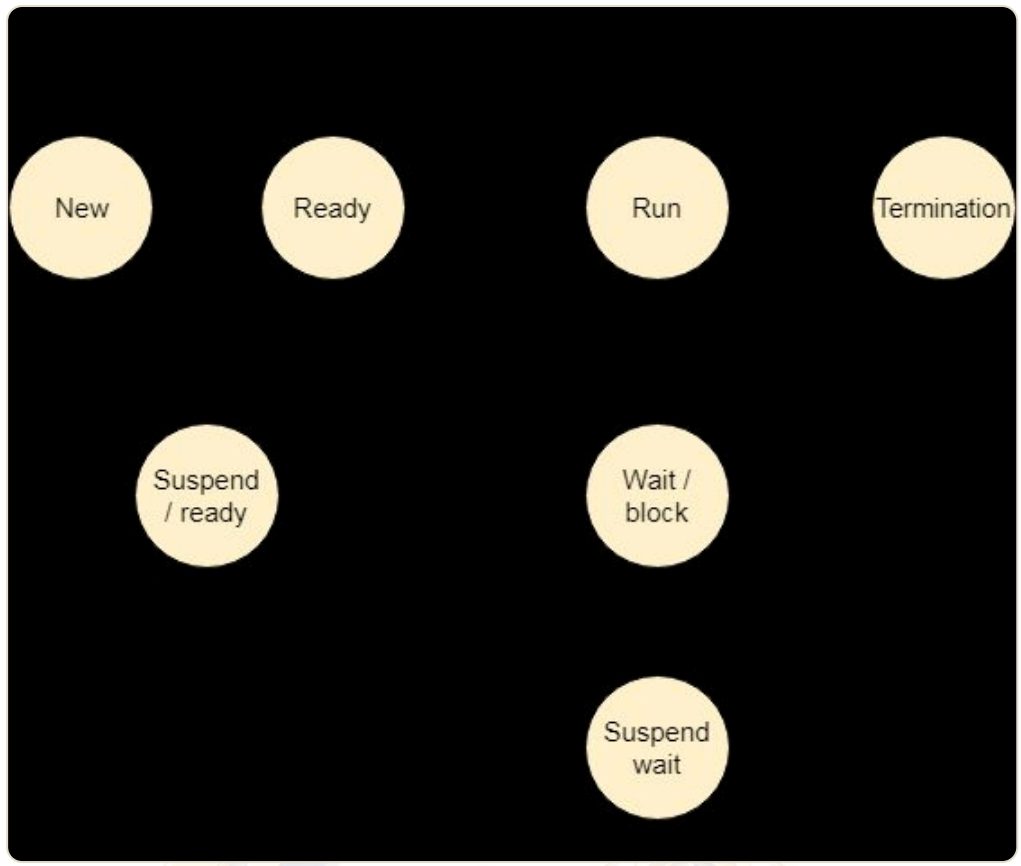
Suspend Wait

|

▼

Waiting

PROCESS STATES



A Process State represents the current status of a process during execution.

1. NEW STATE

When a program is selected by the Operating System and is about to be loaded into main memory, it is in the **New State**.

2. READY STATE

After creation, the process enters the Ready State.

The process is ready to execute but is waiting for CPU allocation.

3. RUNNING STATE

When the CPU starts executing a process, it enters the Running State.

4. BLOCKED / WAITING STATE

A process enters the Waiting State when it needs some event or resource before continuing execution.

Example

A program waiting for data from the hard disk.

5. TERMINATED (COMPLETION) STATE

When a process completes its execution, it enters the Terminated State.

ADDITIONAL PROCESS STATES

Modern Operating Systems also use suspended states.

6. SUSPEND READY STATE

A Ready process moved from main memory to secondary memory due to insufficient memory is called a Suspend Ready Process.

7. SUSPEND WAIT STATE

A Waiting (Blocked) Process moved from main memory to secondary memory is called Suspend Wait State.

Purpose

Frees RAM for other processes.

PROCESS CONTROL BLOCK (PCB)

A PCB (Process Control Block) is a data structure maintained by the Operating System for each process.

OPERATIONS ON A PROCESS

The Operating System performs several operations on processes.

Process Operations

│

├── Creation

├── Scheduling

├── Execution

└── Deletion

1. PROCESS CREATION

The Operating System creates a process and places it in the Ready Queue

2. PROCESS SCHEDULING

Selecting a process from the Ready Queue for execution is called Scheduling.

3. PROCESS EXECUTION

Once selected, the CPU begins executing the process.

Characteristics

- Process enters Running State.

4. PROCESS DELETION (KILLING)

After completion, the Operating System removes the process.

PROCESS SCHEDULING

Process Scheduling is the activity of selecting processes for execution by the CPU.

TYPES OF SCHEDULERS

Schedulers

│

├── Long-Term Scheduler

├── Short-Term Scheduler

└── Medium-Term Scheduler

1. LONG-TERM SCHEDULER

Other Name :Job Scheduler

Selects processes from the job pool (secondary memory) and loads them into the Ready Queue in main memory.

Long-Term Scheduler = Job Scheduler

2. SHORT-TERM SCHEDULER

Other Name :CPU Scheduler

Selects a process from the Ready Queue and assigns CPU to it.

3. MEDIUM-TERM SCHEDULER

The Medium-Term Scheduler manages suspended (swapped-out) processes.

Example

When RAM becomes full, some processes are temporarily moved to disk.

Medium-Term Scheduler = Swapping Manager

COMPARISON OF SCHEDULERS

Feature	Long-Term	Short-Term	Medium-Term
Other Name	Job Scheduler	CPU Scheduler	Swapper
Selects From	Job Pool	Ready Queue	Suspended Processes
Main Function	Load Jobs	Allocate CPU	Swapping
Speed	Slow	Fast	Medium

MOST IMPORTANT FACTS FOR EXAMS

1. Process = Program in Execution.

2. Program is passive; Process is active.

3. PCB stands for Process Control Block.

4. Ready State means waiting for CPU.

5. Running State means CPU is executing.

6. Waiting State means waiting for resource/event.

7. Terminated State means process completed.

8. Suspend Ready = Ready but swapped out.

9. Suspend Wait = Waiting and swapped out.

10. Long-Term Scheduler is also called Job Scheduler.

11. Short-Term Scheduler is also called CPU Scheduler.

12. Medium-Term Scheduler manages swapping.

13. CPU-bound processes require more CPU time.

14. I/O-bound processes require more I/O time.

15. PCB stores process information.

CPU SCHEDULING ALGORITHMS

What is Scheduling?

CPU Scheduling is the process by which the Operating System selects a process from the ready queue and allocates the CPU to it.

Types of Scheduling Algorithms

1. Preemptive Scheduling

2. Non-Preemptive Scheduling

A. PREEMPTIVE SCHEDULING

1. Shortest Remaining Time First (SRTF)

- CPU is allocated to the process having the **shortest remaining execution time**.
- If a new process arrives with a shorter remaining time, the currently running process is interrupted.

2. Priority Scheduling (Preemptive)

- Each process is assigned a priority.
- Higher-priority process gets CPU first.

3. Round Robin (RR)

- Most commonly used in time-sharing systems.
- Each process gets a fixed time slice called **Time Quantum**.
- After the quantum expires, the process returns to the ready queue.

4. Longest Remaining Time First (LRTF)

- Process with the longest remaining execution time gets CPU first.

B. NON-PREEMPTIVE SCHEDULING

1. First Come First Serve (FCFS)

- Process arriving first gets CPU first.

2. Shortest Job First (SJF)

- Process with shortest burst time executes first

3. Longest Job First (LJF)

- Process with longest burst time executes first..

PROCESS RELATED TIMES

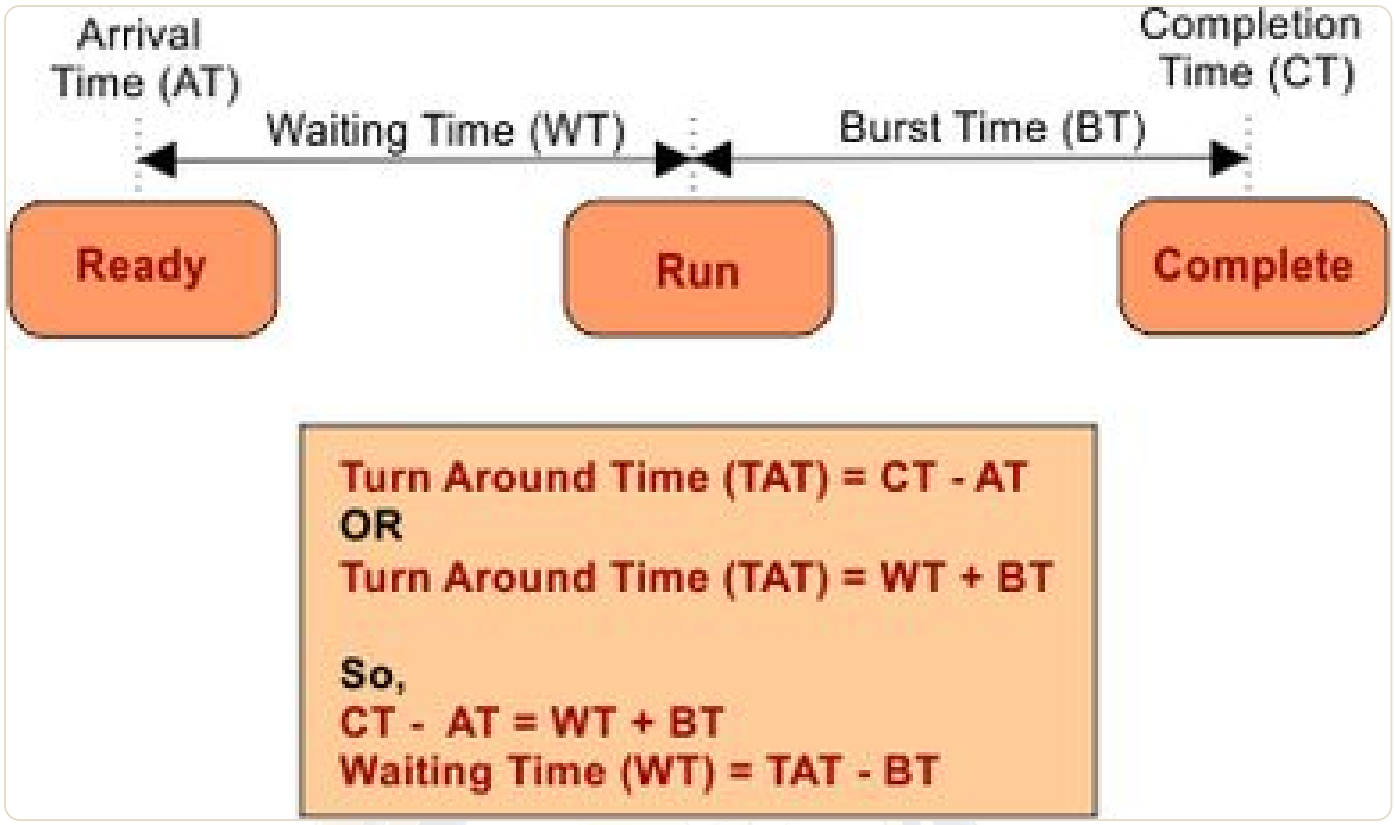
1. Arrival Time (AT)

Time at which a process enters the ready queue.

Example

Process arrives at 10:00 AM.

Arrival Time = 10:00 AM



2. Burst Time (BT)

Amount of CPU time required to execute the process.

Example

Process requires 5 seconds of CPU.

Burst Time = 5 sec

Exam Point

- ✔ Does not include waiting time.

3. Completion Time (CT)

Time at which process finishes execution.

4. Turnaround Time (TAT)

Total time taken from arrival to completion.

5. Waiting Time (WT)

Time spent waiting in the ready queue.

6. Response Time (RT)

Time between arrival and first CPU allocation.

PROCESS SYNCHRONIZATION

Process Synchronization ensures that multiple cooperating processes execute in the correct order and share resources safely.

Why Needed?

When multiple processes access the same data simultaneously, incorrect results may occur.

Example

Two persons withdrawing money from the same bank account at the same time.

SYNCHRONIZATION MECHANISMS

Critical Section: the part of a program where a shared resource is accessed. Only one process should be in the critical section at a time (Mutual Exclusion).

- Semaphore – an integer variable used to control access to shared resources using wait() (P) and signal() (V) operations. Introduced by Dijkstra.
- Mutex (Mutual Exclusion Lock) – a lock that allows only one process/thread to enter the critical section at a time.
- Monitor – a high-level synchronization construct in which only one process can be active inside the monitor at a time.

FILE SYSTEMS

FAT (File Allocation Table)

- Older file system used in Windows.
- Simple structure.

NTFS (New Technology File System)

- Modern Windows file system.
- Supports:
  - Security
  - Encryption
  - Large files
  - Permissions

Comparison

FAT	NTFS
Older	Newer
Less Secure	More Secure
Small File Support	Large File Support
No Permissions	Supports Permissions

BOOTING

Booting is the process of starting a computer and loading the operating system into memory.

Booting Sequence

Power ON

↓

BIOS

↓

POST

↓

Boot Loader

↓

Operating System

↓

User Login

BIOS (Basic Input Output System)

- Stored in ROM.
- Initializes hardware.

POST (Power-On Self-Test)

- Checks RAM, keyboard, CPU, etc.

Boot Loader

Loads the operating system from secondary storage (HDD/SSD) into RAM.

Examples

- GRUB (Linux)
- Windows Boot Manager

Bootstrap Loader

- Stored in ROM.
- Loads BIOS/initial boot program into memory.
- Starts the booting process

HIBERNATION AND SLEEP MODE

Hibernation

- Current state of system is saved to hard disk.
- Power is completely turned off.
- On restart, previous session is restored.

Sleep Mode

- Current state stored in RAM.
- Very little power consumed.
- Faster resume than hibernation.

DEADLOCK

A deadlock is a situation where two or more processes are waiting for each other indefinitely, and none can proceed.

Example

Process P1 holds Resource R1 and waits for R2.

Process P2 holds Resource R2 and waits for R1.

Result: Deadlock

Real-Life Example

Two cars stuck on a narrow bridge facing each other.

Four Necessary Conditions for Deadlock (Coffman Conditions)

- Mutual Exclusion – a resource can be used by only one process at a time.
- Hold and Wait – a process holds one resource and waits for another.
- No Preemption – a resource cannot be forcibly taken from a process.
- Circular Wait – processes wait for each other in a circular chain.

Methods of Deadlock Handling

1. Deadlock Prevention

Prevent one of the four conditions from occurring.

2. Deadlock Avoidance

System avoids entering unsafe state.

Example

Banker's Algorithm

3. Deadlock Detection and Recovery

Allow deadlock to occur, detect it, then recover.

4. Ignore Deadlock

Known as Ostrich Approach.

System assumes deadlock is rare.

Exam Point

✔ Most general-purpose OS use the Ostrich Approach.

OPEN SOURCE OPERATING SYSTEM

An Open Source Operating System is an OS whose source code is publicly available and can be viewed, modified, and distributed by users.

Advantages

- Free to use
- Customizable
- Community support
- Secure due to open inspection

Examples

- Linux

- Ubuntu
- Fedora
- Debian
- Android

FAMOUS OPERATING SYSTEMS

Operating system names	Developer	RUN On which devices
MS-DOS(disk operating system)	Microsoft (1981), command based OS	PC
Microsoft Windows	Microsoft (1985), graphical OS	PC
Mac OS (MAcintosh)	Apple (1984; Mac OS X in 2001)	PC
Linux	Linus torvalds	Server(96 percent of server used this),phone, supercomputer
Android	Open handset alliance(2008)	Mobile phones
IOS	Apple (2007)	Mobile phones
UNIX	Ken,dennis(1970)	Servers
Ubuntu	Mark shuttleworth	Cloud computing ,internet of devices,computer
Solaris	Sun microsystem	Database OS

MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

1. SRTF is the preemptive version of SJF.

2. Round Robin uses Time Quantum.

3. SJF gives minimum average waiting time.

4. Bootstrap Loader resides in ROM.

5. Boot Loader loads OS into RAM.

6. NTFS is more secure than FAT.

7. Sleep mode stores data in RAM.

8. Hibernation stores data on Hard Disk.

9. Banker's Algorithm is used for Deadlock Avoidance.

10. Deadlock requires all four Coffman conditions

11. Linux is the most popular open-source operating system.